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WATERWAY INSTALLATION STRUCTURE IN A TOILET

Abstract

The application discloses a waterway installation structure in a toilet, comprising at least two waterway modules and a hard connection module for connecting two the waterway modules; each of the waterway modules comprises a waterway joint; the hard connection module comprises a module body having a watercourse and at least two connection pipe joints provided on the module body, each of the connection fittings is in communication with the watercourse; each of the waterway joints on the waterway module is connected and sealed with one connection pipe joint; the waterway joint is plugged into the connection pipe joint; or, the connection pipe joint is plugged into the waterway joint. The waterway installation structure in the toilet disclosed by the application is easy to assemble, and conducive to saving installation space and improving the structural stability.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/244,033 (Docket No. 010222-23043A) filed on Sep. 8, 2023, which is hereby incorporated by reference in its entirety, and which claims the benefit of priority to Chinese Patent Application No. 202222448019.5 filed in the Chinese Intellectual Property Office on Sep. 15, 2022, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates to the technical field of toilet, and particularly relates to a waterway installation structure in a toilet.

BACKGROUND

[0003] Many waterway modules may be mounted in a back base of a smart toilet, for example, a water purification module, a spray pipe module, a waterway switching module, etc. These modules need to be connected to each other through a water pipe in order to achieve waterway connection.

[0004] The waterway modules that need to be connected to each other are connected through a silicone hose. A certain amount of interference should be left when a silicone hose is cooperated or engaged with a joint, and the hose needs to be used in conjunction with a clamp in the process of assembly to achieve a sealed connection.

[0005] Generally, installation spaces of the waterway modules are small, and the user's hand is difficult to reach into the installation space to conduct the installation. The installation method by using the hose and the clamp is time-consuming and laborious, and the working hour is long. A sufficient bending space needs to be reserved at the connection of the hose to accommodate extra hoses. This takes up a lot of space.

SUMMARY

[0006] The present disclosure aims to overcome the above defects and provides a novelty waterway installation structure in a toilet, which comprises a connection module that is easy to be assembled and takes up small space.

[0007] An embodiment of the present disclosure provides a waterway installation structure in a toilet, comprising at least two waterway modules and a connection module (e.g., a hard connection module or a connection module made from hard material) configured to be connected to the two waterway modules.

[0008] Each of the waterway modules comprises a waterway joint.

[0009] The connection module comprises a module body comprising a watercourse and comprises at least two connection pipe joints disposed on the module body. Each of the connection pipe joints is in communication with (e.g., in fluid communication with) the watercourse.

[0010] The waterway joint of each waterway module is connected to and sealed with one connection pipe joint of the at least two connection pipe joints.

[0011] The waterway joint is plugged (e.g., inserted) into the connection pipe joint, or the

connection pipe joint is plugged (e.g., inserted) into the waterway joint.

[0012] In another embodiment of the present disclosure, a seal ring is disposed between the waterway joint and the connection pipe joint.

[0013] In another embodiment of the present disclosure, the waterway joint or the connection pipe joint comprises a limit groove, and the seal ring is assembled or disposed in the limit groove.

[0014] In another embodiment of the present disclosure, the connection pipe joint is formed integrally with the module body.

[0015] In another embodiment of the present disclosure, the module body is provided with a body jaw disposed on a side of each connection pipe joint,

[0016] The waterway module comprises a limit block disposed on a side of each waterway joint.

[0017] The body jaw is in snap-fit connection with the limit block.

[0018] In another embodiment of the present disclosure, the waterway installation structure further comprises a structural base, and the structural base comprises a mounting groove configured to mount the module body.

[0019] The module body is removably mounted in the mounting groove.

[0020] In another embodiment of the present disclosure, the structural base is provided with at least one positioning pin around the mounting groove. Correspondingly, the module body is provided with at least one positioning lug plate.

[0021] Each of the at least one positioning lug plate comprises a positioning hole configured to cooperate with the at least one positioning pin.

[0022] The positioning lug plate is disposed on an outside of the mounting groove, and each of the at least one positioning pin passes through one corresponding positioning hole with a clearance therebetween.

[0023] In another embodiment of the present disclosure, the mounting groove is provided with the at least one positioning pin disposed on each of opposite sides of the mounting groove.

Correspondingly, the module body is provided with the at least one positioning lug plate disposed on each of opposite sides of the module body.

[0024] In another embodiment of the present disclosure, the structural base is provided with at least one base jaw around the mounting groove, and the base jaw is in snap-fit connection with the module body.

[0025] In another embodiment of the present disclosure, the mounting groove is provided with the at least one base jaw disposed on each of opposite sides of the mounting groove. Each of the base jaws is in snap-fit connection with the module body.

[0026] By adopting the technical solutions above, the present disclosure has the following beneficial effects:

[0027] According to the waterway installation structure in a toilet provided by the present disclosure, the connection module is adopted, the waterway module comprises the waterway joint, and the connection module comprises the connection pipe joint. When the waterway installation structure is connected or assembled, the waterway joint may be directly inserted into the connection pipe joint to achieve a sealed connection, or the connection pipe joint may be directly inserted into the waterway joint to achieve a sealed connection. It is easy to assemble the waterway installation structure, and thus time and effort may be saved. The connection module does not take up extra installation space but takes up small space. This is convenient for structural arrangement. The connection module plays a structural connection role between the two waterway modules and helps to improve the stability of the connection between the two waterway modules.

[0028] In summary, the waterway installation structure in the toilet provided by the present disclosure is easy to be assembled, saves the installation space, and also improves the structural stability.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0029] With reference to the drawings, the contents disclosed by the present disclosure should be more easily understood. It should be understood that these drawings are merely used for illustration and are not intended to limit the protection scope of the present disclosure. In the drawings:

[0030] FIG. 1 is a perspective view of a waterway installation structure in the toilet according to an example of the present disclosure;

[0031] FIG. 2 is an exploded view of the waterway installation structure in the toilet shown in FIG. 1;

[0032] FIG. 3 is a perspective view of a waterway module of a structure;

[0033] FIG. 4 is a perspective view of a waterway module of another structure;

[0034] FIG. 5 is a perspective view of a connection module;

[0035] FIG. 6 is a sectional view of a module body with a watercourse;

[0036] FIG. 7 is a perspective view of the connection module connected to a structural base;

[0037] FIG. 8 is an exploded view of the connection module connected to the structural base;

[0038] FIG. 9 is a partially enlarged view of the structural base;

[0039] FIG. 10 is an enlarged view of the connection module connected to the structural base; and

[0040] FIG. 11 is a flow chart of a method for assembling the waterway installation structure in the toilet according to an example of the present disclosure.

DETAILED DESCRIPTION

[0041] The specific embodiments of the present disclosure are further described with reference to the drawings hereinafter. Same or equivalent parts are denoted by same reference numerals. It should be noted that the terms “front”, “back”, “left”, “right”, “up” and “down” used in the following description refer to the directions in the drawings, and the terms “inner” and “outer” refer to the directions towards or far away from geometric centers of specific parts respectively.

[0042] FIG. 1 is a perspective view of a waterway installation structure in the toilet according to an example of the present disclosure. FIG. 2 is an exploded view of the waterway installation structure in the toilet shown in FIG. 1. FIG. 3 is a perspective view of a waterway module of a structure. FIG. 4 is a perspective view of a waterway module of another structure. FIG. 5 is a perspective view of a connection module. FIG. 6 is a sectional view of a module body with a watercourse. As shown in FIGS. 1-6, a waterway installation structure in a toilet according to an embodiment of the present disclosure comprises at least two waterway modules 1 and a hard connection module 2 configured to be connected to the at least two waterway modules 1.

[0043] Each waterway module 1 comprises a waterway joint 11.

[0044] The hard connection module 2 comprises a module body 21 comprising a watercourse 211 and comprises at least two connection pipe joints 22 disposed on the module body 21. Each of the connection pipe joints 22 is in communication with (e.g., in fluid communication with) the watercourse 211.

[0045] The waterway joint 11 on each waterway module 1 is connected to and sealed with one connection pipe joint 22.

[0046] The waterway joint 11 is plugged (e.g., inserted) into the connection pipe joint 22, or the connection pipe joint 22 is plugged (e.g., inserted) into the waterway joint 11.

[0047] The present disclosure provides a waterway installation structure mainly used in the toilet. Specifically, the waterway installation structure may be mounted in a back base of the toilet or a water tank. The back base used in this present disclosure may be a mounting base of a toilet cover.

[0048] The waterway installation structure comprises the waterway module 1 and the hard connection module 2. The waterway module 1 may be a water-using device, such as a motor, a water purifier, a spray pipe, etc. The waterway module 1 may also be a water-passing device, such

as a water valve, pipe fitting, etc. The waterway module **1** may also be a combination of the water-using devices and the water-passing devices.

[0049] The waterway installation structure comprises at least two waterway modules **1**. Each waterway module **1** comprises a waterway joint **11**. The waterway joint **11** may be an inlet or outlet of the waterway module **1**, and specific settings may be made according to actual needs. When the at least two waterway modules **1** are connected to a connection module **2** (e.g., a hard connection module or a connection module made from hard material). One waterway joint **11** of one of the at least two waterway modules **1** is an outlet of the corresponding waterway module **1**, a waterway joint **11** of the other waterway module **1** of the at least two waterway modules **1** is an inlet of the corresponding waterway module **1**.

[0050] When two waterway modules **1** need to achieve waterway connection, the hard connection module **2** may be used for the connection.

[0051] The hard connection module **2** may be a plastic connection module or a metal connection module. Compared to the rubber hose, the plastic connection module or the metal connection module is harder in texture and will not bend. Thus, the hard connection module **2** may be called "hard connection module."

[0052] The hard connection module **2** comprises a module body **21** and at least two connection pipe joints **22**. As needed, the module body **21** may be configured in an elongated shape with a watercourse **211** inside. The number of the connection pipe joints **22** is configured according to the number of waterway joints **11** to be connected. The at least two connection pipe joints **22** are connected to the module body **21**, and each connection pipe joint **22** is in communication with (e.g., in fluid communication with) the watercourse **211**.

[0053] In this present disclosure, both the waterway joint **11** and the connection pipe joint **22** are cylindrical.

[0054] According to actual needs, an outer diameter of the waterway joint **11** may be set to be slightly smaller than an inner diameter of the connection pipe joint **22**, so that the waterway joint **11** can be directly plugged into the connection pipe joint **22** to achieve the connection.

[0055] According to actual needs, alternatively the outer diameter of the connection pipe joint **22** may be set to be slightly smaller than the inner diameter of the waterway joint **11**, so that the connection pipe joint **22** can be directly plugged into the waterway joint **11** to achieve the connection.

[0056] When the waterway installation structure is assembled, the waterway joint **11** on each waterway module **1** may be directly plugged into the corresponding connection pipe joint **22** to achieve a sealed connection, or each connection pipe joint **22** of the hard connection module **2** may be plugged into the corresponding waterway joint **11** to achieve a sealed connection.

[0057] As needed, a tapered sealing surface is provided on surfaces (e.g., mating surfaces) of the at least two waterway joints **11** and the at least two connection pipe joints **22** to achieve automatic sealing after the waterway joint **11** is plugged into the connection pipe joint **22** or the connection pipe joint **22** is plugged into the waterway joint **11**.

[0058] It is also possible to coat sealant onto the mating surfaces of the waterway joint **11** and the connection pipe joint **22**, as needed, to achieve automatic sealing after the waterway joint **11** is plugged into the connection pipe joint **22** or the connection pipe joint **22** is plugged into the waterway joint **11**.

[0059] The waterway installation structure in the toilet provided in the present disclosure adopts the hard connection module **2**. When the waterway installation structure is connected or assembled, the waterway joint **11** may be directly inserted into the connection pipe joint **22** to achieve a sealed connection, or the connection pipe joint **22** may be directly inserted into the waterway joint **11** to achieve a sealed connection. It is easy to assemble the waterway installation structure, and thus time and effort may be saved. The hard connection module **2** does not take up extra installation space but takes up small space. This is convenient for structural arrangement. The hard connection

module **2** plays a structural connection role between the two waterway modules **1** and helps to improve the stability of the connection between the two waterway modules **1**.

[0060] In summary, the waterway installation structure in the toilet provided by the present disclosure is easy to be assembled, saves the installation space, and also improves the structural stability.

[0061] In an embodiment, as shown in FIG. **2**, a seal ring **3** is disposed between the waterway joint **11** and the connection pipe joint **22** to further improve sealing performance. The seal ring **3** is a rubber seal ring.

[0062] In an embodiment, as shown in FIGS. **3** and **4**, the waterway joint **11** or the connection pipe joint **22** comprises a limit groove **4**, and the seal ring **3** is assembled or disposed in the limit groove **4** to improve the assembly stability of the seal ring **3**.

[0063] If the waterway joint **11** is plugged into the connection pipe joint **22**, the limit groove **4** is configured or disposed on an outer periphery surface of the waterway joint **11**.

[0064] If the connection pipe joint **22** is plugged into the waterway joint **11**, the restriction groove **4** is configured or disposed on an outer peripheral surface of the connection pipe joint **22**.

[0065] In an embodiment, the connection pipe joint **22** is formed integrally with the module body **21**, resulting in high structural strength. If the hard connection module **2** is a hard plastic component, the connection pipe joint **22** is formed integrally with the module body **21** by injection molding. If the hard connection module **2** is a metal component, the connection pipe joint **22** is formed integrally with the module body **21** by casting.

[0066] In an embodiment, as shown in FIGS. **1-5**, the module body **21** is provided with a body jaw **23** disposed on a side of each connection pipe joint **22**.

[0067] The waterway module **1** comprises a limit block **12** disposed on a side of the waterway joint **11**.

[0068] The body jaw **23** is in snap-fit connection with the limit block **12**.

[0069] In this embodiment, the hard connection module **2** comprises the body jaw **23** to provide a limit to the module body **21**. The body jaw **23** is connected to the module body **21**, and the body jaw **23** and the module body **21** may be formed integrally. The body jaw **23** extends from the module body **21** toward the waterway module **1**.

[0070] Specifically, one body jaw **23** is disposed on the side of each connection pipe joint **22**.

[0071] Each waterway module **1** comprises one limit block **12**, and the limit block **12** extends horizontally or laterally. The limit block **12** may be formed integrally with the waterway module **1**. The limit block **12** is disposed on the side of the waterway joint **11**.

[0072] When the waterway installation structure is assembled, each body jaw **23** on the hard connection module **2** is snapped onto limit block **12** of one waterway module **1** to improve the assembly stability of the hard connection module **2** and the waterway module **1**.

[0073] In an embodiment, the hard connection module **2** is disposed under or below the waterway module **1**, and the body jaw **23** is disposed on a top surface of the module body **21**.

[0074] Specifically, the body jaw **23** comprises a standing plate **231** connected to the module body **21** and a hanging plate **232** connected to a top of the standing plate **231** and extending toward a side of the standing plate **231**. The hanging plate **232** is hung on the limit block **12** when the hard connection module **2** is assembled to the waterway module **1**.

[0075] As needed, a top surface of the hanging plate **232** and a bottom surface of the limit block **12** each may be configured with or comprise a corresponding bevel. When the hard connection module **2** is assembled with the waterway module **1**, the standing plate **231** is deformed under the action of the bevel, so that the hanging plate **232** may move onto the limit block **12**, and thus the hanging plate **232** may automatically hang on the limit block **12** after the reset of the standing plate **231**.

[0076] FIG. **7** is a perspective view of the connection module connected to a structural base. FIG. **8** is an exploded view of the connection module connected to the structural base. FIG. **9** is a partially

enlarged view of the structural base. FIG. 10 is an enlarged view of the connection module connected to the structural base. In an embodiment, as shown in FIGS. 7-10, the waterway installation structure also comprises a structural base 5 comprising a mounting groove 51 configured to mount the module body 21. The module body 21 is removably mounted in the mounting groove 51.

[0077] In this embodiment, the waterway installation structure comprises the structural base 5, and the structural base 5 may be a pedestal of a back base of a toilet or toilet bidet.

[0078] The structural base 5 has the mounting groove 51, which matches the shape contour of the module body 21, and the module body 21 is removably mounted in the mounting groove 51, enhancing the installation stability of the hard connection module 2.

[0079] In this present disclosure, the at least two hard connection modules 2 have installation positions at a top surface of the structural base 5, and the module body 21 is installed in the mounting groove 51 of the structural base 5 to further improve the stability of the waterway installation structure.

[0080] In an embodiment, as shown in FIGS. 7-10, at least one positioning pin 52 is disposed configured on the structural base 5 around the mounting groove 51. Correspondingly, at least one positioning lug plate 24 is disposed on the module body 21.

[0081] Each of the positioning lug plates 24 comprises a positioning hole 241 configured to cooperate with (e.g., receive) the positioning pin 52.

[0082] The positioning lug plate 24 is disposed on an outside of the mounting groove 51, and each of the positioning pins 52 passes through the one positioning hole 241 with a clearance therebetween.

[0083] In this embodiment, at least one positioning pin 52 is disposed on the structural base 5, and the positioning pin 52 is disposed on the outside of the mounting groove 51. The module body 21 is correspondingly provided with at least one positioning lug plate 24. The positioning lug plate 24 extends laterally from the module body 21, which may be formed integrally with the module body 21. Each of the positioning lugs 24 has at least one positioning hole 241, which is used to cooperate with the positioning pin 52 to achieve the positioning.

[0084] When the waterway installation structure is assembled, the positioning lug plate 24 is disposed on the outside of the mounting groove 51, and the positioning lug plate 24 covers the positioning pin 52. Each positioning pin 52 passes through and is in clearance fit with one positioning hole 241.

[0085] In an embodiment, as shown in FIGS. 7-10, at least one positioning pin 52 is disposed on each of the opposite sides of the mounting groove 51 respectively. Correspondingly, at least one positioning lug plate 24 is disposed on each of the opposite sides of the module body 21 respectively.

[0086] In this embodiment, the positioning pin 52 is disposed on each side of the long side of the mounting groove 51 respectively. Correspondingly, the positioning lug plate 24 is disposed on each of the two long sides of the module body 21 respectively. When the waterway installation structure is assembled, the positioning lug plate 24 is disposed on the outside of the mounting groove 51, and the positioning lug plate 24 covers the positioning pin 52. Each positioning pin 52 cooperates with the positioning hole 241 on the corresponding positioning lug plate 24 to achieve the positioning and improve the structural stability.

[0087] In an embodiment, as shown in FIGS. 7-10, the structural base 5 is provided with at least one base jaw 53 around the mounting groove 51, and the base jaw 53 is in snap-fit connection with the module body 21.

[0088] In this embodiment, the base jaw 53 is provided on the structural base 5, and the base jaw 53 is on the outside of the mounting groove 51, and the mounting groove 51 may also be provided directly on a groove wall of the mounting groove 51. The base jaw 53 may have a similar structural design to the body jaw 23.

[0089] After the module body **21** is assembled into the mounting groove **51**, the snap plate or hanging plate of the base jaw **53** is snapped onto the top surface of the module body **21**. Thus, the module body **21** is limited in the mounting groove **51**, and the assembly stability of the module body **21** to the mounting groove **51** may be improved.

[0090] In an embodiment, as shown in FIGS. **7-10**, the mounting groove **51** comprises at least one base jaw **53** disposed on each opposite sides respectively, and each of the base jaws **53** may be snapped onto the module body **21**.

[0091] In this embodiment, the base jaw **53** is provided on each side of the long side of the mounting groove **51**. During the assembly, the base jaws **53** on both sides are snapped onto the module body **21**, and thus the assembly stability of the module body **21** may be further improved.

[0092] In an embodiment, the toilet includes a base (e.g., a pedestal, bowl, etc.) and a tank. The base is configured to be attached to another object such as a drainpipe, floor, or another suitable object. The base includes a bowl, a sump (e.g., a receptacle) disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The tank may be supported by the base, such as an upper surface of a rim. The tank may be integrally formed with the base as a single unitary body. In other embodiments, the tank may be formed separately from the base and coupled (e.g., attached, secured, fastened, connected, etc.) to the base. The toilet may further include a tank lid covering an opening and inner cavity in the tank. The toilet may include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilet may further include a hinge assembly.

[0093] In another embodiment, the toilet may be a tankless toilet. The toilet includes a base and a seat assembly coupled to the base. The base includes a bowl, a sump disposed below the bowl, and a trapway fluidly connecting the bowl to a drainpipe or sewage line. The toilet includes a waterline that supplies the toilet with water. The toilet may further include a seat assembly including a seat and a seat cover rotatably coupled to the base. The toilets described above are provided herein as non-limiting examples of toilets that may be configured to utilize aspects of the present disclosure.

[0094] In some examples, a bidet may be included in a seat or pedestal of a toilet. In other examples, the bidet may be manufactured separately from and attached or coupled to a seat or pedestal of a toilet. The bidet includes a housing. The housing is configured to receive a flow of water through a housing inlet and dispense the flow of water from a housing outlet. The housing inlet and housing outlet may be located on opposite ends of the housing from one another, such that water may flow through the housing from the housing inlet to the housing outlet. In some examples, the housing further includes a chamber. As the housing receives the flow of water, the chamber may fill with water and provide a flow of water between the housing inlet and the housing outlet. The chamber may be configured to contain the flow of water and direct the flow of water from the housing inlet to the housing outlet. After the chamber has filled with water, the flow of water may travel along a substantially linear path between the housing inlet and the housing outlet. In some examples, one or more walls within the housing may be included to help direct a flow of water between the housing inlet and the housing outlet. The bidet may further include a housing inlet conduit configured to direct a flow of water to the housing inlet. The housing inlet conduit may be coupled to a water supply such as tank or waterline. The housing may further include a gear assembly or a portion of the gear assembly.

[0095] FIG. **11** is a flow chart of a method for assembling the waterway installation structure in the toilet according to an example of the present disclosure. The the waterway installation structure in the toilet used in the method may be the waterway installation structure in the toilet according to any of the foregoing examples and may be configured to perform an operation, function, or the like as described in the present disclosure.

[0096] At act **S101**, the at least two waterway joints **11** of the at least two waterway modules **1** are respectively connected to the at least two connection pipe joints **22** of the connection module **2**.

[0097] As noted above, the waterway joint **11** is inserted into the corresponding connection pipe

joint **22**, or the connection pipe joint **22** is inserted into the corresponding waterway joint **11**. The seal ring **3** is disposed between the waterway joint **11** and the connection pipe joint **22** to further improve sealing performance. The seal ring **3** is a rubber seal ring. The waterway joint **11** or the connection pipe joint **22** comprises a limit groove **4**, and the seal ring **3** is assembled or disposed in the limit groove **4** to improve the assembly stability of the seal ring **3**. If the waterway joint **11** is inserted into the connection pipe joint **22**, the limit groove **4** is configured or disposed on an outer periphery surface of the waterway joint **11**. If the connection pipe joint **22** is inserted into the waterway joint **11**, the restriction groove **4** is configured or disposed on an outer peripheral surface of the connection pipe joint **22**.

[0098] At act **S102**, the at least two body jaws **23** disposed on a top surface of the module body **21** of the connecting module **2** are snapped onto the at least two limit blocks **12** disposed on the at least two waterway joints **11**.

[0099] As noted above, when the waterway installation structure is assembled, each body jaw **23** on the hard connection module **2** is snapped onto limit block **12** of one waterway module **1** to improve the assembly stability of the hard connection module **2** and the waterway module **1**. The body jaw **23** comprises a standing plate **231** connected to the module body **21** and a hanging plate **232** connected to a top of the standing plate **231** and extending toward a side of the standing plate **231**. The hanging plate **232** is hung on the limit block **12** when the hard connection module **2** is assembled to the waterway module **1**.

[0100] At act **S103**, the at least two waterway joints **11** are disposed on the top surface of the module body **21**.

[0101] As noted above, the hard connection module **2** is disposed under or below the waterway module **1**, and the body jaw **23** is disposed on a top surface of the module body **21**. As needed, a top surface of the hanging plate **232** and a bottom surface of the limit block **12** each may be configured with or comprise a corresponding bevel. When the hard connection module **2** is assembled with the waterway module **1**, the standing plate **231** is deformed under the action of the bevel, so that the hanging plate **232** may move onto the limit block **12**, and thus the hanging plate **232** may automatically hang on the limit block **12** after the reset of the standing plate **231**.

[0102] The above technical solutions may be combined as required to achieve the best technical effect.

[0103] The above are merely the principle and the preferred embodiments of the present disclosure. It should be pointed out that, for those of ordinary skills in the art, several other modifications may be made on the basis of the principle of the present disclosure, which should also be regarded as falling in the protection scope of the present disclosure.

Claims

1. A waterway installation structure in a toilet, the waterway installation structure comprising: a connection module mounted to the toilet, the connection module comprising: a module body including a watercourse and a connection pipe joint in fluid communication with the watercourse; and at least one waterway module including a waterway joint configured to be connected to the connection pipe joint, wherein the waterway joint is configured to be inserted into the connection pipe joint or the connection pipe joint is configured to be inserted into the waterway joint.
2. The waterway installation structure of claim 1, further comprising: a seal ring disposed between the waterway joint and the connection pipe joint.
3. The waterway installation structure of claim 2, further comprising: a limit groove disposed on the waterway joint or the connection pipe joint, the seal ring disposed in the limit groove.
4. The waterway installation structure of claim 1, wherein the connection pipe joint is formed integrally with the module body.
5. The waterway installation structure of claim 1, further comprising: a body jaw disposed on the

module body on a side of the connection pipe joint; and a limit block disposed on the waterway joint, wherein the body jaw is configured to be connected to the limit block.

6. The waterway installation structure of claim 5, wherein: the body jaw includes a standing plate extending from the module body, and a hanging plate connected to the standing plate; the body jaw is disposed on a top surface of the module body; and the water module is disposed on the top surface of the module body.

7. The waterway installation structure in the toilet according to claim 1, further comprising: a structural base comprising a mounting groove, wherein the module body is removably mounted in the mounting groove.

8. The waterway installation structure in the toilet according to claim 7, the waterway installation structure further comprising: at least one positioning pin disposed on the structural base around the mounting groove, wherein the module body includes at least one positioning lug plate, wherein the at least one positioning lug plate includes at least one positioning hole configured to receive the at least one positioning pin, and wherein the positioning lug plate is disposed on an outside of the mounting groove when the at least one positioning pin passes through the at least one positioning hole.

9. The waterway installation structure in the toilet according to claim 8, wherein: the at least one positioning pin comprises two positioning pins respectively disposed on opposite sides of the mounting groove, and the at least one positioning lug plate comprises two positioning lug plates respectively disposed on opposite sides of the module body.

10. The waterway installation structure in the toilet according to claim 7, further comprising: at least one base jaw disposed on the structural base and around the mounting groove, the at least one base jaw configured to be connected to the module body.

11. The waterway installation structure in the toilet according to claim 10, wherein the at least one base jaw comprises two base jaws respectively on opposite sides, the two base jaws configured to be connected to the module body.

12. A method for assembling a waterway installation structure, the method comprising: connecting at least two waterway joints of at least two waterway modules to respective ones of at least two connection pipe joints of a connection module; disposing the at least two waterway joints on a top surface of the connection module; and mounting the waterway installation structure to a toilet.

13. The method of claim 12, further comprising: connecting a seal ring between a waterway joint of the at least two waterway joints and a corresponding connection pipe joint of the at least two connection pipe joints.

14. The method of claim 12, wherein connecting the at least two waterway joints of the at least two waterway modules to respective ones of the at least two connection pipe joints of the connection module includes: inserting the at least two waterway joints into respective ones of the at least two connection pipe joints.

15. The method of claim 12, wherein connecting the at least two waterway joints of the at least two waterway modules to respective ones of the at least two connection pipe joints of the connection module includes: inserting the at least two connection pipe joints into respective ones of the at least two waterway joints.

16. The method of claim 12, further comprising: disposing a module body including the at least two connection pipe joints in a mounting groove of a structural base.

17. The method of claim 16, further comprising: inserting at least one positioning pin disposed on the structural base into at least one positioning hole of at least one positioning lug plate of the module body.

18. A waterway installation structure in a toilet, the waterway installation structure comprising: a connection module mounted to the toilet, the connection module comprising: a module body including a watercourse and a connection pipe joint in fluid communication with the watercourse; and at least one waterway module including a waterway joint configured to be connected to the

connection pipe joint; and a structural base including a mounting groove, the mounting groove configured to receive the module body, wherein the waterway joint is configured to be inserted into the connection pipe joint or the connection pipe joint is configured to be inserted into the waterway joint.

19. The waterway installation structure of claim 18, further comprising: a seal ring disposed between the waterway joint and the connection pipe joint; and a limit groove disposed on the waterway joint or the connection pipe joint, wherein the seal ring disposed in the limit groove.

20. The waterway installation structure of claim 18, further comprising: a body jaw disposed on the module body on a side of the connection pipe joint; and a limit block disposed on the waterway joint, wherein the body jaw is configured to be connected to the limit block.
